

YUROU CELINE XIAO

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Skills

Hardware Design & Manufacturing: high-pressure (10 bar) high-temperature (100°C) cell design, pilot scaling (800 cm²/cell, 10 cells/stack), SolidWorks, engineering drawings, GD&T, laser cutter, drill press, soldering, electronics assembly

Electrochemistry & Fabrication: LANDT Battery Test Systems, Metrohm Autolab, CC/CV, LSV, EIS, catalyst synthesis, electrode fabrication, electrolyte formulation

Analytical Chemistry & Software: SEM, XRD, GC, IC, NMR, UV-vis, Python, MATLAB, OriginPro

Professional Experience

Program Lead | SintonLab, University of Toronto | Toronto, ON May 2023 – August 2025

- Directed a multi-year, \$2M+ research program sponsored by Shell International B.V. and the Government of Canada focused on integrated capture and electrochemical conversion of CO₂.
- Led a team of up to 5 researchers to achieve energy-efficient (>40%) and durable (>1,000 hr) CO₂ electrolysis, resulting in multiple publications in high-impact peer-reviewed journals.

Electrolyzer Fabrication Coordinator | SintonLab, University of Toronto | Toronto, ON September 2023 – August 2025

- Served as the primary technical liaison between researchers and machinists, translating complex electrolyzer design requirements into manufacturable components while ensuring structural integrity and functionality.

Doctoral Researcher | CSIRO Energy Center | Newcastle, Australia January – April 2024

- Developed and scaled CO₂ electrolyzers from 1 cm² to 100 cm² for integrated CO₂ capture and conversion systems.

Co-founder | E-quester | Toronto, ON September 2021 – October 2022

- Co-founded E-quester, a startup team developing a low-cost and scalable electrically powered direct air capture (DAC) system for CO₂ removal, resulting in a \$250,000 Student Prize and named a Top 60 Milestone Award Finalist in the XPRIZE Carbon Removal competition.

Pilot Operator | CERT Systems Inc. | Calgary, AB September – December 2020

- Scaled the electrochemical conversion of CO₂ into ethylene by operating a pilot plant (up to 40,000 cm² total active area), contributing to Team CERT's success as a Top 10 Finalist in the NRG COSIA Carbon XPRIZE.
- Operated and maintained a multi-electrolyzer pilot, ensuring continuous system performance through assembly, troubleshooting, and dynamic response to upstream process upsets.

Project Engineering Intern | WSP in Canada | Calgary, AB August – December 2019

- Developed a proprietary Class V capital cost estimation tool in Excel for industrial processing facilities, leveraging historical data and scalable process equipment models to improve budget forecasting accuracy.
- Transformed legacy P&IDs into a 3D-linked compliance tool and authored core engineering documentation (DBM, PSV/DIB lists) to ensure adherence to Canadian methane regulations.

Project Management Intern | City of Edmonton | Edmonton, AB May – December 2018

- Acted as lead delegate in site meetings for municipal infrastructure projects; owned issue resolution by authorizing change orders and coordinating with cross-functional teams to maintain project milestones.

Equipment Reliability Engineering Intern | Syncrude | Fort McMurray, AB September – December 2017

- Engineered and delivered reliability packages to resolve both emergency equipment failures and scheduled maintenance, ensuring full compliance with ASME B31.3, B31.1, Sec. VIII Div. 1, and API 650 codes.

Education

Doctor of Philosophy in Mechanical and Industrial Engineering September 2020 – October 2025

SintonLab, University of Toronto

Toronto, ON

- Thesis: *Electrochemical Systems for Reactive Capture of CO₂*. Supervised by Prof. David Sinton

Bachelor of Science in Mechanical Engineering

University of Alberta

September 2015 – April 2020

Edmonton, AB